

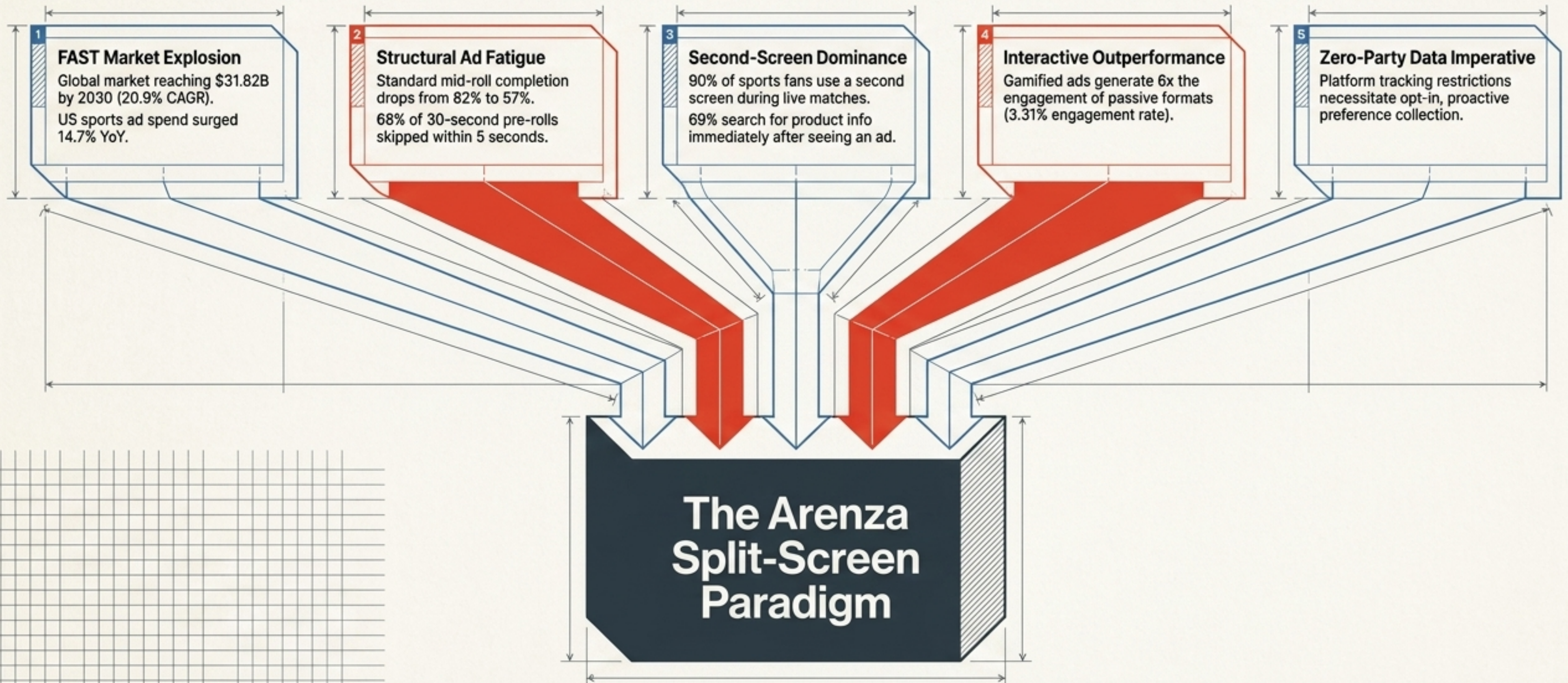
The Arenza Architecture Blueprint

**Engineering the
Intersection of Sports
FAST Streaming,
Gamification, and Local
SMB Offline Attribution.**

GRID REF: 48.15N / 11.58E

STRUCTURAL INTEGRITY LEVEL: MAX

The Inevitable Convergence Point



The Split-Screen Paradigm

Top Layer: Content

Low-latency live Sports FAST feed. Immune to ad blockers, capitalizing on the \$5.3B AVOD sports content market.



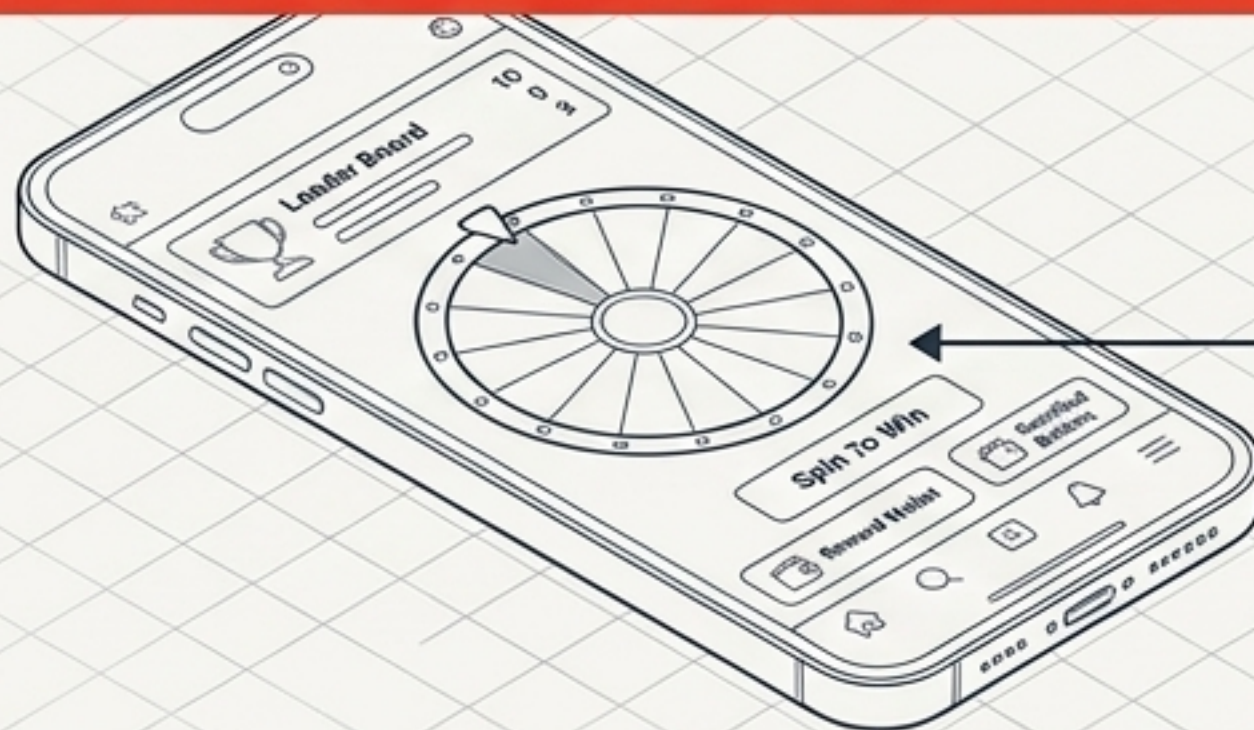
Zero-interruption viewing

42.7% of viewers use ad-blocking; 57% of mid-rolls exceeding 20s are skipped.

Core Advantage: Value is presented alongside content, completely bypassing the structural failure modes of passive video advertising.

Bottom Layer: Commerce & Data

Synchronized, opt-in interactive zone. Server-authoritative gamification replacing standard SSAI mid-rolls.



Real-time game synchronization

Industry Validation: Circling the Solution

Amazon + DraftKings / FanDuel

Integrating live odds and custom content overlays directly into TNF broadcasts.

StreamLayer

Proving overlay interactivity lifts in-app session minutes by 2-3x on game days.

NBCUniversal

Deploying Arrival Ads and pause-ad programmatic buying for Peacock live sports.

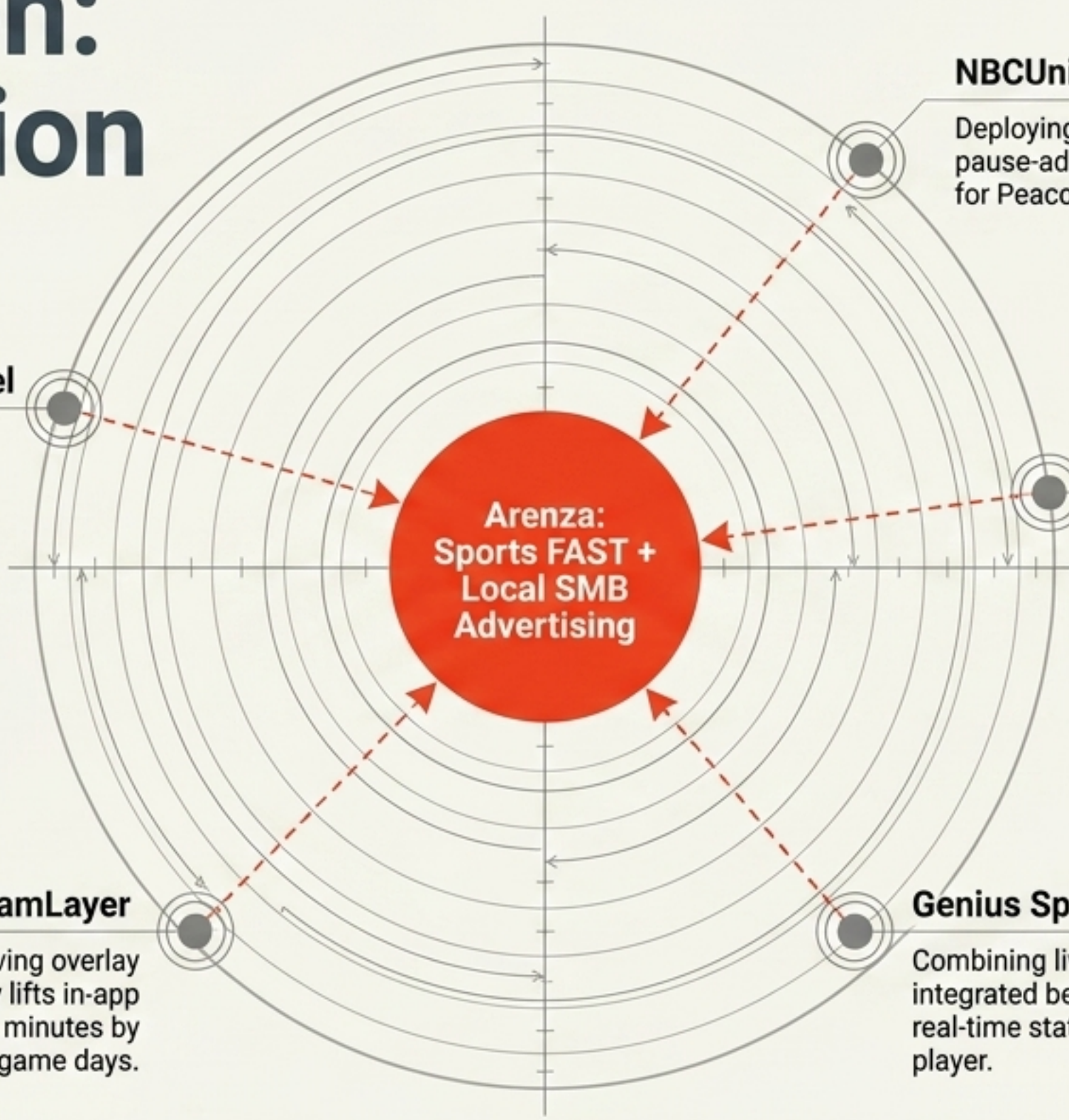
Roku Ads Manager

Launching shoppable CTV campaigns with QR overlays as first-class interactive formats.

Genius Sports BetVision

Combining live video with integrated bet slips and real-time stats in a single player.

Arenza:
Sports FAST +
Local SMB
Advertising



The White Space Diagnostic

Arenza's intersection of sports FAST streaming, gamified mini-games, local geo-targeting, and POS-linked redemption represents a multi-billion dollar whitespace untouched by national incumbents.

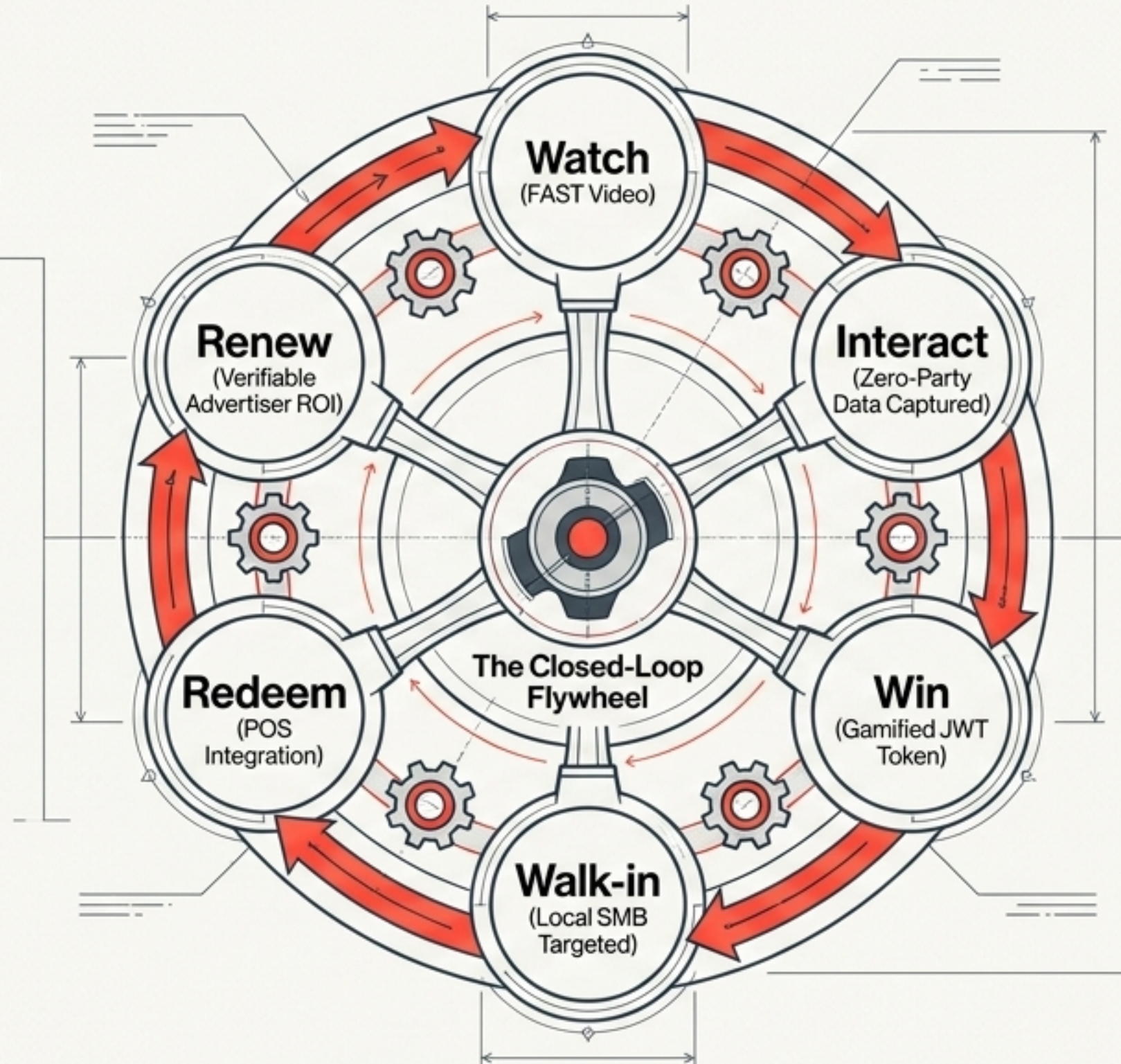
Platform	Video Layer	Interactive Layer	Local SMB Targeting	QR Attribution	Offline Redemption
Arenza	Sports FAST stream	Gamified spin/scratch	 Location + preference	 POS-linked	 Physical reward claim
Roku Ads Manager	CTV/streaming	QR overlays, shoppable	Zip-level geo	Partial	[X]
StreamLayer	Sports live stream	Polls, predictions, stats	[X]	[X]	[X]
BrightLine	CTV	Remote-interactive games	[X]	[X]	[X]
Genius BetVision	Sports live stream	Betting integration	[X]	[X]	[X]
DraftKings x Amazon	NFL/NBA streams	Odds, parlays	[X]	[X]	[X]
Tubi	FAST broad	Standard pre/mid-roll	Regional data	[X]	[X]

The Twin Engines of Value: Data & Attribution

Zero-Party Data Engine

The on-device ProfileEngine uses CoreML to process watch-time patterns and explicit game interactions.

Voluntarily shared data circumvents Apple's App Tracking Transparency friction while maintaining total privacy compliance.



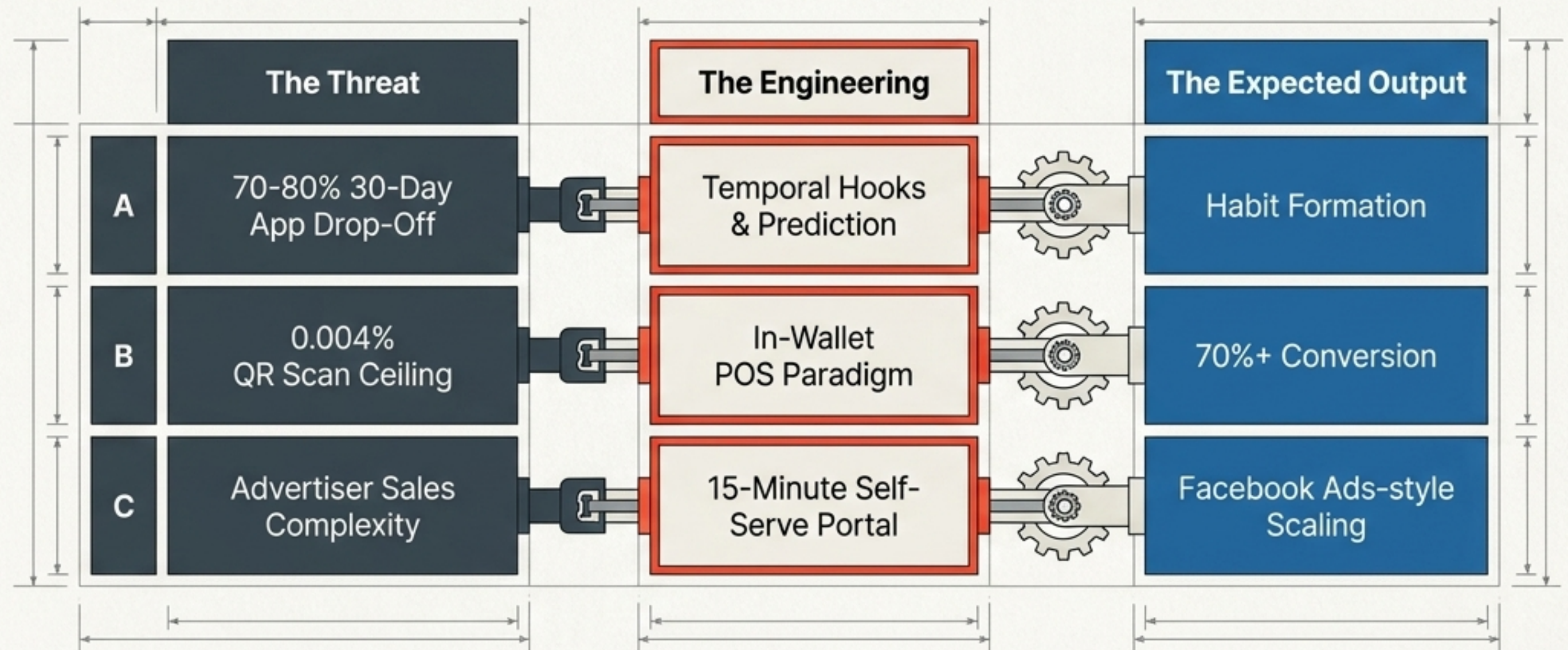
The Holy Grail of Attribution

Connects the digital impression directly to the physical point-of-sale.

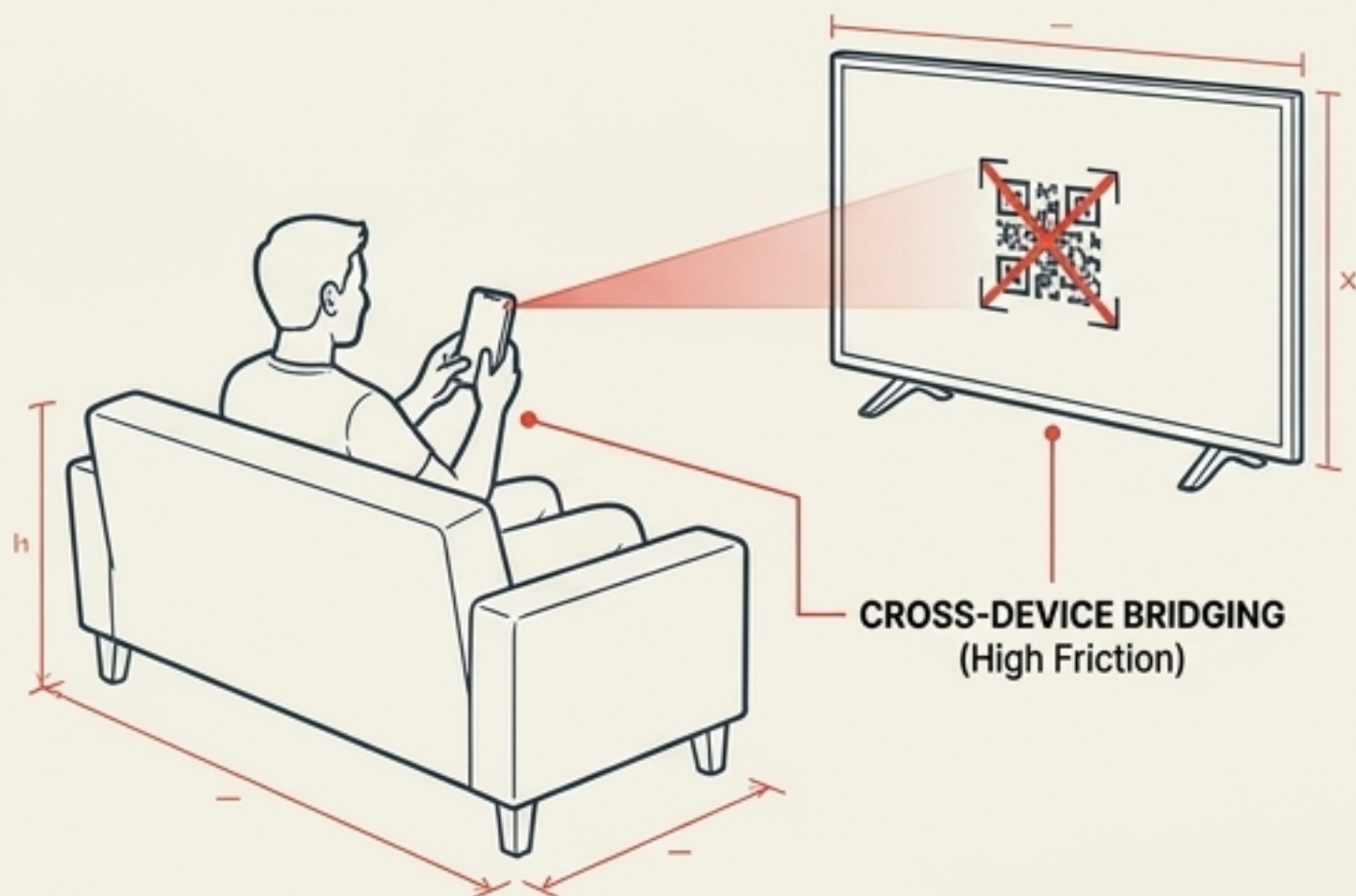
Benchmark Validation:
Foursquare CTV attribution studies confirm exposed users drive a 2.93% conversion rate and a 5% purchase intent increase.

Confronting the Friction: Structural Risk Mitigation

We acknowledge three critical failure points inherent to companion apps and interactive CTV formats, and deploy programmatic, structural countermeasures to completely neutralize them.



The QR Paradigm Shift



The Misconception

BrightLine data shows a 0.004% median scan rate for CTV campaigns (400 scans per 10M impressions). This data measures large-screen-to-mobile scanning, a high-friction action requiring users to pick up a phone and point it at a distant TV.

Cross-Device Bridging
(High Friction = 0.004% success)



The Reality

Arenza utilizes **in-app wallet-to-POS** presentation (low friction, phone already in hand). Planters' Super Bowl QR drove a 70% conversion rate. Heavy streamers are 2x more likely to scan. This paradigm reflects consumer readiness for low-friction digital.

In-Wallet Point-of-Sale
(Low Friction = 70% Conversion)

Engineering Habit: The Temporal Retention Loop

The Challenge:

Companion apps notoriously lose **70-80%** of users within 30 days due to post-game irrelevance.

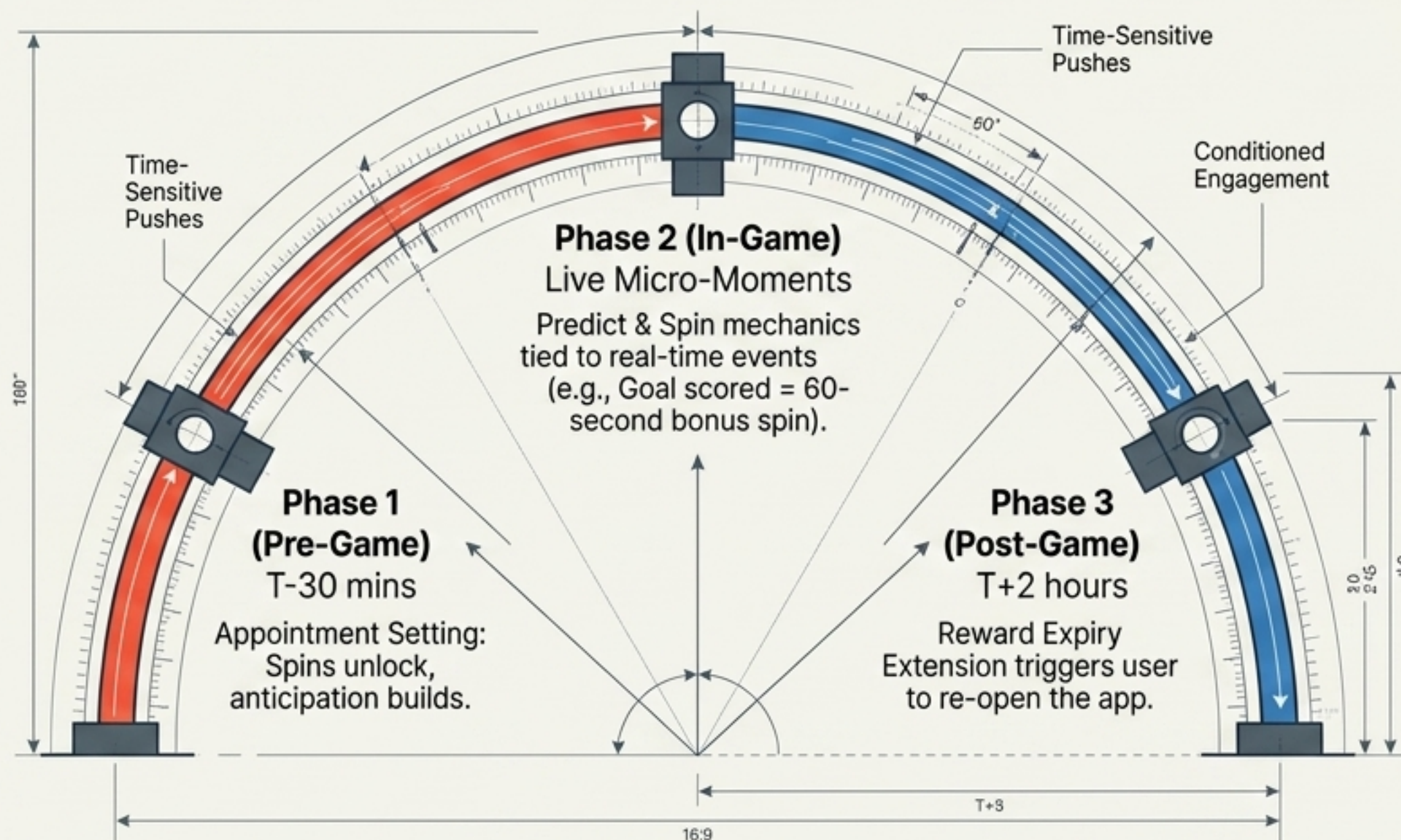
The Fix:

Structuring engagement as an event-driven loop to drive **20-40%** higher session times and **15-30%** higher retention.

Push Re-Engagement:

83% of Americans use mobile while watching TV. Live-game time-sensitive pushes yield **3-7x** normal conversion rates.

The Game Day Timeline



The Four-Layer Technical Architecture

Layer 1: Streaming Infrastructure

HLS/DASH ingestion, adaptive bitrate CDN (Cloudflare/AWS), Server-Side Ad Insertion (SSAI) suppression. Target latency: <500ms end-to-end.

Layer 2: On-Device ProfileEngine

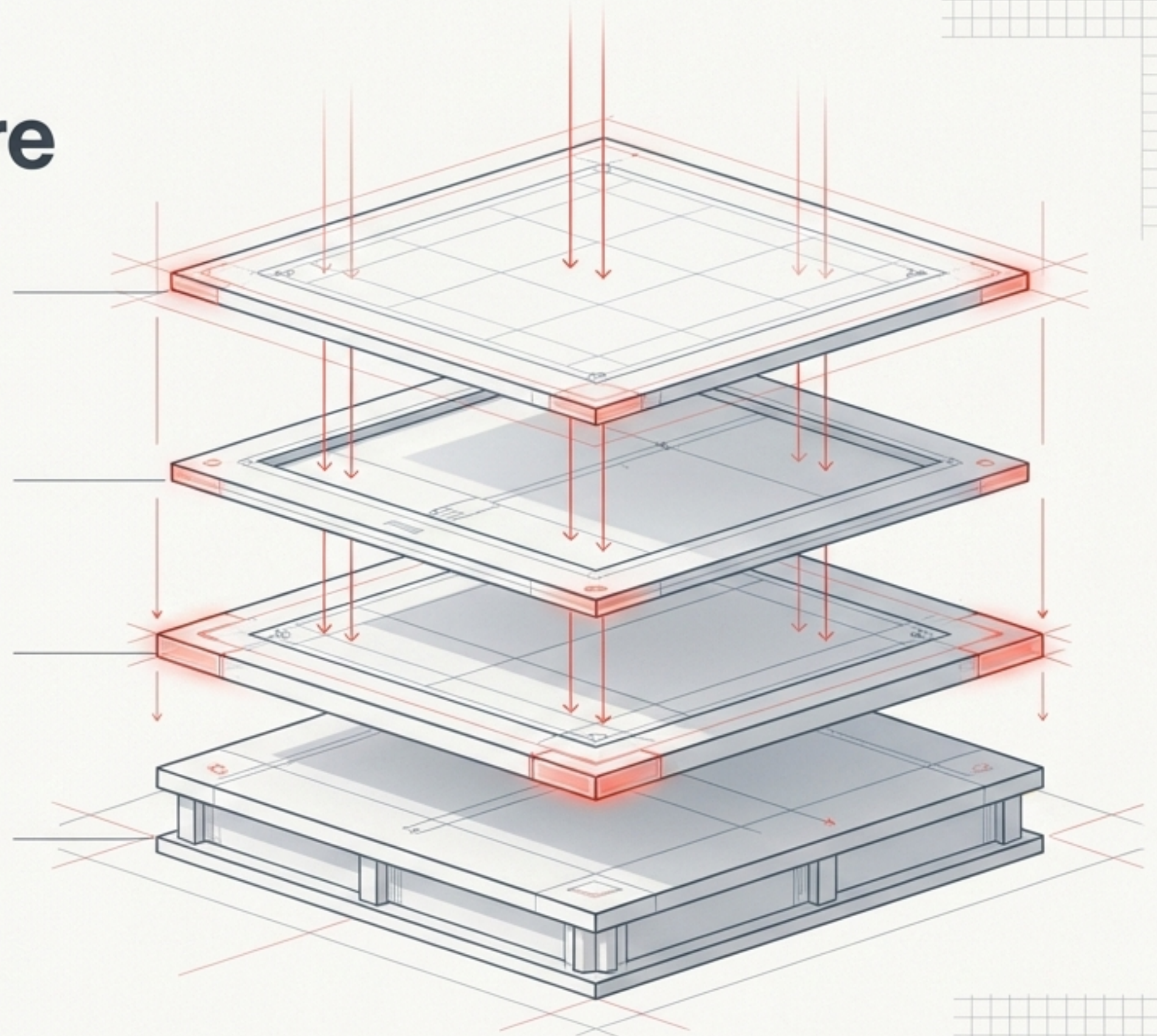
CoreML watch-pattern processing, foreground location services for 5-mile geo-targeting.

Layer 3: Game & Reward Engine

Server-authoritative mechanics (zero client-side manipulation), HMAC-SHA256 signed JWT tokens, Redis session state, WebSockets.

Layer 4: POS Integration

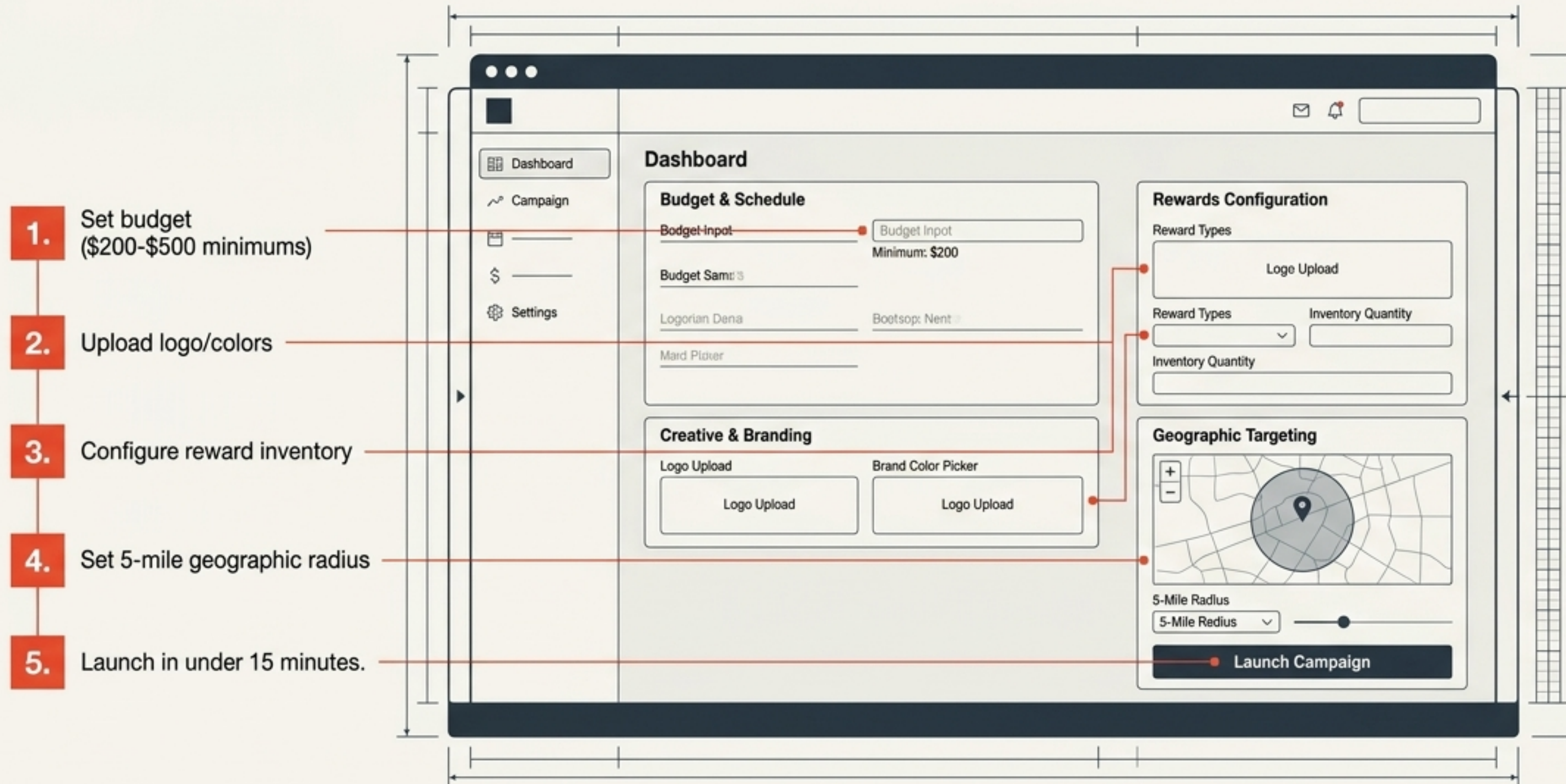
Phase 1 iOS operator scanner via AVFoundation, evolving to REST webhooks (Toast, Square, Clover).



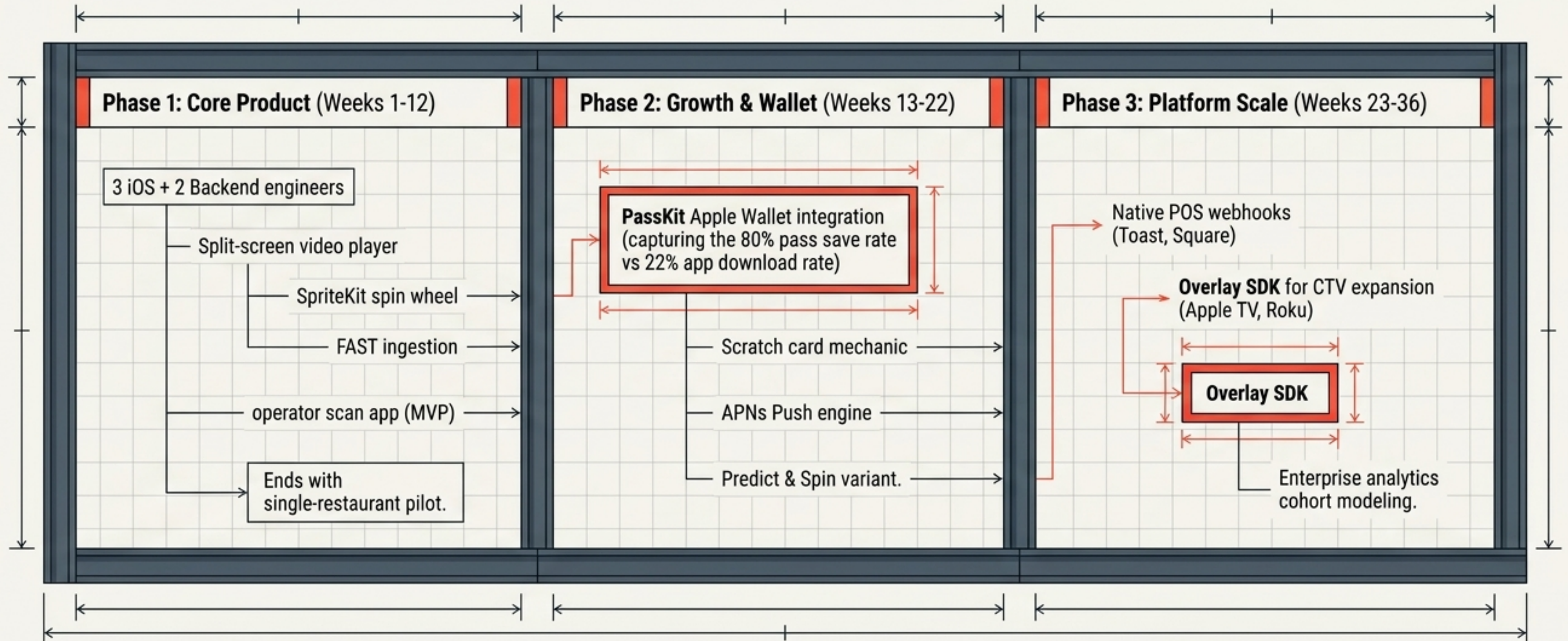
Scaling the Whitespace: The Self-Serve Advertiser Portal

The Bottleneck: Local SMBs cannot navigate complex programmatic CTV buying or month-long POS API integrations.

The Precedent: Directly mirrors Roku Ads Manager's successful democratization of CTV for non-technical buyers.



The 36-Week Rollout Blueprint



The Unit Economics Waterfall

Four Revenue Streams:

1. Base SaaS subscription
(\$79-\$299/mo)
2. CPM performance fee
(\$3-\$8 per 1k impressions)
3. Redemption fee
(\$0.05-\$0.15 per claim)
4. Data insights tier

300 Daily Active Viewers x 2.5 spins x 30 days
= 22,500 Game Sessions

35% Win Rate

7,875 Rewards Generated

75% Claim Rate

5,906 POS Physical Redemptions
(\$590 redemption revenue)

**Gross Revenue: \$199 (Base Sub) + \$112 (CPM) + \$590 (Redemption)
= ~\$901/month per restaurant.**

Performance Trajectories & KPI Benchmarks

KPI Metric	Baseline Target	Mature Target
Daily Active Users (DAU/MAU)	>20%	>40%
Spins per session	1.8	2.5+
Reward claim rate	>60%	>80%
QR-to-POS scan completion	>45%	>65%
30-day user retention	>35%	>55%
Advertiser renewal rate	>60%	>80%
Session duration vs. non-interactive	+40%	+100%

The Durable Strategic Moat

Arenza is the definitive closed-loop operating system for local CTV advertising. By flawlessly executing non-interruptive delivery, zero-party data collection, and highly frictionless POS attribution, Arenza has built an infrastructure impervious to ad blockers, immune to standard engagement fatigue, and insulated from FAST market incumbents.

The ultimate wedge: Bridging the \$12.28B FAST ecosystem directly to the local SMB cash register.